

2(a)	<i>first experiment:</i> rate of oxygen production ; <i>second experiment:</i> time taken to change colour (from blue-green to green) / rate of DCPIP reduction ;	2 ignore rate of photosynthesis allow volume / number of moles / AW, of oxygen produced, in a set time ignore rate of NADP reduction
2(b)	<i>any one from:</i> 1 mass / concentration, of <i>A. doliolum</i> (cells in suspension) ; 2 concentration of enzyme ; 3 time, in enzyme solution / for hydrolysis of cell walls ; 4 concentration of DCPIP ;	1 ignore light intensity / temperature ignore volume (of solutions) ignore ref to colour ignore pH ignore carbon dioxide ignore stirring R concentration of sodium chloride 1 allow <i>A. doliolum</i> / cells / <i>Anabaena</i> / prokaryote as AW for <i>A. doliolum</i> cells ignore size 3 ignore time unqualified R time taken to change colour
2(c)	<i>idea that</i> (enzyme partially hydrolysed the bacterial cell walls so): (more) DCPIP can, diffuse / enter, cells or diffusion / AW, of DCPIP (into cells) is, faster / AW or permeability of cell <u>wall</u> increases ;	1 ignore <i>ref. to</i> sodium chloride / NaCl ignore <i>ref. to</i> NADP / H⁺ ignore enter chloroplast ignore easier

Question	Answer	Marks	Guidance																								
2(d)	<table><tr><td colspan="3">Table 2.1 completed with correct volumes ;</td></tr><tr><td>concentration of sodium chloride solution / moldm⁻³</td><td>volume of stock solution of sodium chloride / cm³</td><td>(volume) water ; (/ cm³)</td></tr><tr><td>0.25</td><td>10</td><td>10</td></tr><tr><td>0.20</td><td>8</td><td>12</td></tr><tr><td>0.15</td><td>6</td><td>14</td></tr><tr><td>0.10</td><td>4</td><td>16</td></tr><tr><td>0.05</td><td>2</td><td>18</td></tr><tr><td>0.00</td><td>0</td><td>20</td></tr></table>	Table 2.1 completed with correct volumes ;			concentration of sodium chloride solution / moldm ⁻³	volume of stock solution of sodium chloride / cm ³	(volume) water ; (/ cm ³)	0.25	10	10	0.20	8	12	0.15	6	14	0.10	4	16	0.05	2	18	0.00	0	20	2	ignore decimals ignore working / $C_1V_1 = C_2V_2$
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2(e)	1 highest / AW, (mean rate of) oxygen production is at a <u>lower sodium chloride concentration</u> than the highest AW, (mean rate of) DCPIP reduction ; ora 2 highest / AW, (mean rate of) oxygen production is at 0.10 mol dm⁻³ (sodium chloride) and highest / optimum / AW, (mean rate of) DCPIP reduction is at 0.20 mol dm⁻³ (sodium chloride) ;	2	ignore rate of photosynthesis unqualified ignore first / second experiment allow NaCl as AW for sodium chloride allow peak / optimum / greatest / maximum as AW for highest ignore any comparison between y-axis values ignore y-axis data quotes ignore <i>ref. to</i> lines crossing ignore comparisons between gradients of lines (e.g. fast decrease / slight decrease) 2 need units for concentration at least once